

Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CIT 320 : Software Development Project-II

Project Report



Project Title : RentSyne(Flat Rent and Service Charge Management System)

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RentSyne

1. Introduction

In urban apartment systems, flat rent collection and service charge management are often time-consuming, error-prone, and manually handled. This creates confusion, delays, and lack of transparency between flat owners and tenants. To address these issues, the **Flat Rent and Service Charge Management System** is proposed — a smart, digital platform to streamline rent tracking, payment management, and service billing for residential buildings.

2. Objectives

The main objective of this project is to develop a mobile-based system that allows:

1. Easy and secure rent & service charge management
2. Transparent payment tracking
3. Digital communication between owner and tenant
4. Automated receipt generation and reminders

3. Problem Statement

Managing flat rent and monthly service charges in apartment buildings is often a manual, disorganized, and inefficient process. Flat owners or building managers typically rely on paper records, Excel sheets, or personal communication (calls/messages) to collect payments, maintain billing history, and notify tenants — which leads to:

1. Miscommunication between tenants and owners
2. Delays in rent or service charge collection
3. Lack of proper tracking for paid/unpaid bills
4. Difficulty generating receipts and reports
5. Inability to send timely reminders or updates

4. Related Work

1. **RentRedi** — A U.S.-based rent management app for landlords and tenants. Provides rent collection, tenant screening, and maintenance tracking.

Limitation: Mainly targeted at large-scale property managers in the U.S. and not customizable for local needs (e.g., Bangladesh). Also, no free version available.[1]

2. **Buidium** — A powerful property management system with rent tracking, accounting, and maintenance requests.

Limitation: Expensive, overkill for small apartment buildings. Requires a web browser; not optimized for mobile-first users.

3. **TenantCloud** — Cloud-based rental management platform that allows online payments and document storage.

Limitation: Too feature-heavy and complex for basic flat rent tracking in small apartments. Lacks localized support.

4. **BariBodol (Bangladesh)** — A local platform for flat rental listings and communication between owners and tenants.

Limitation: Focused only on finding flats. It does not manage rent collection, service charges, or tenant history.[2]

5. **Manual Systems (Paper/Excel/Calls)** — Most flat owners in South Asia still use notebooks, Excel sheets, or messaging apps to collect rent.

Limitation: Prone to human error, lacks automation, no digital receipts, no historical tracking, and inefficient communication.

5. Scope

The Flat Rent and Service Charge Management System will be developed as a cross-platform mobile application using Flutter, allowing seamless access on both Android and iOS devices. The primary goal is to digitalize and streamline rent collection, service charge tracking, and tenant management for small to mid-sized apartment buildings.

This application aims to be simple, scalable, and customizable, making it ideal for individual building owners, housing societies, or real estate agencies.

6. Methodology

The development of the Flat Rent and Service Charge Management System will follow an Agile approach using Iterative and Incremental Development, enabling rapid prototyping, continuous feedback, and flexible improvements based on user testing.

6.1. Step-by-Step Development Methodology

1. Requirement Analysis

- I. Meet with stakeholders (flat owners, tenants) to gather system requirements
- II. Identify core modules (rent tracking, user roles, payment, receipts)
- III. Finalize functional and non-functional requirements
- IV. Define user stories and use-case scenarios

2. System Design

- I. Design system architecture (frontend, backend, database)
- II. Create UI/UX wireframes and screen flows in Figma
- III. Define database schema for tenants, flats, rent, charges, and payments
- IV. Plan role-based access control for Admin and Tenant

3. Technology Stack Selection

- I. Frontend: **Flutter** (Android & iOS support)
- II. Backend: **Firestore** (or **PHP + MySQL** depending on scale)
- III. Database: **Firestore**
- IV. Authentication: **Firestore Auth**
- V. Notifications: **Firestore Cloud Messaging (FCM)**
- VI. PDF Generation: Dart packages like pdf, printing

Notes

- I. Iterations ship incremental features for early user feedback
- II. Continuous testing ensures stability across mobile platforms
- III. Security practices include secure authentication, data validation, and audit logs

7. Visual Models

7.1. Flow Chart Diagram

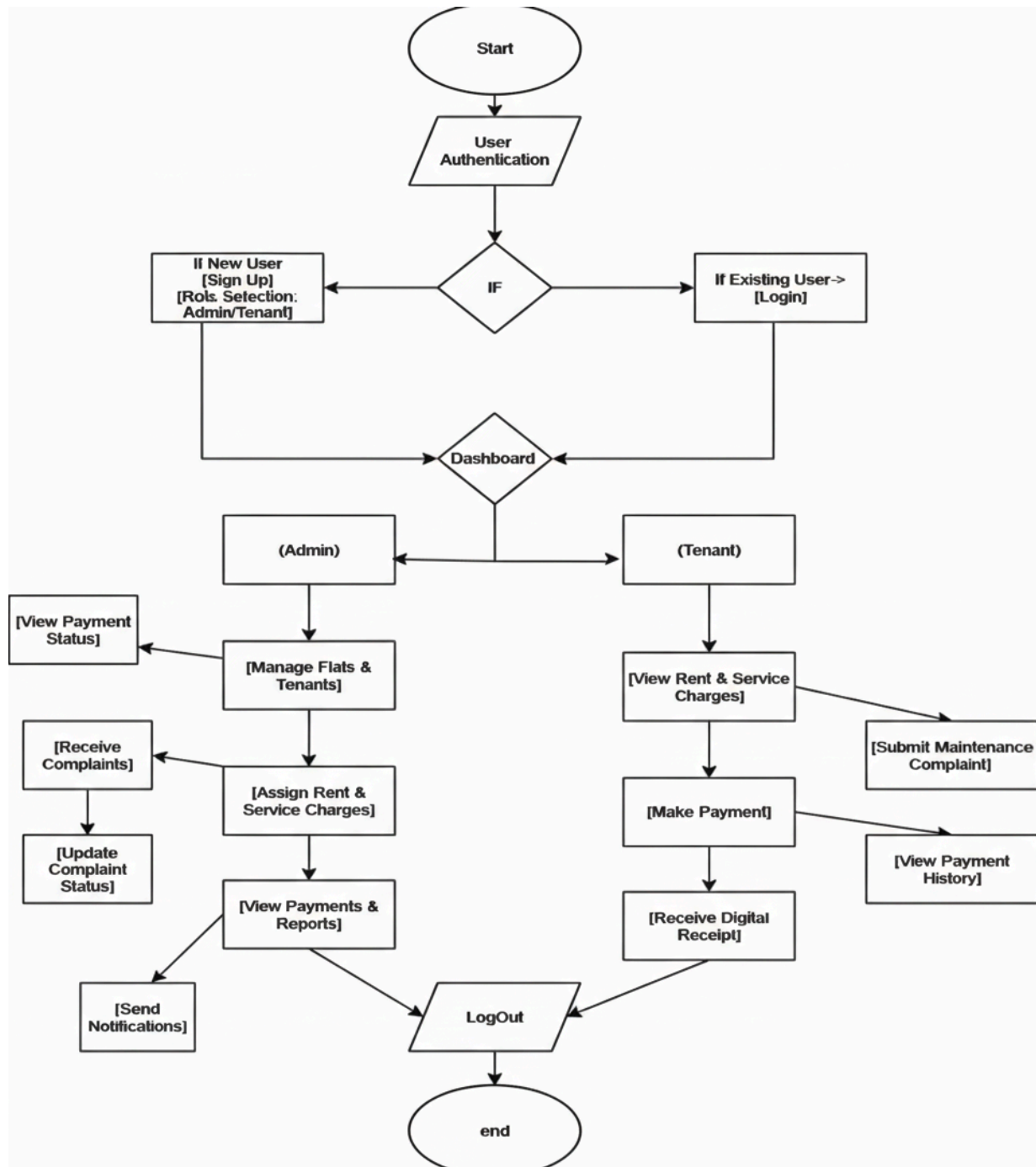


Figure 1: Flow Chart of RentSyne Architecture

The above Figure 1 illustrates the overall architecture of the RentSyne application, showing the interaction between users, the frontend application, backend services, and the database. It highlights the flow of data and the key components involved in the system.

7.2. ERD (Entry Relationship Diagram)

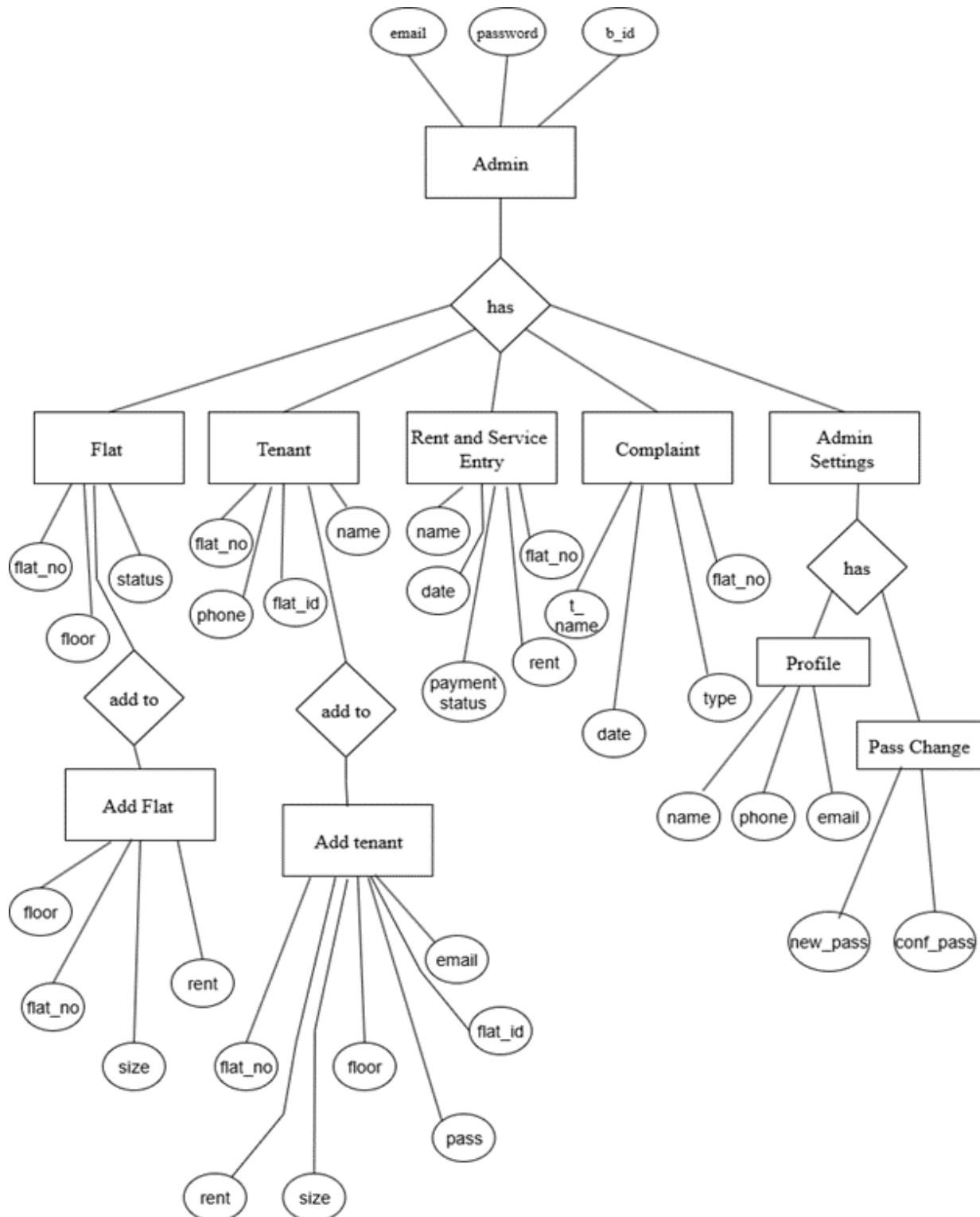


Figure 2: Entity Relationship Diagram of RentSyne

The above Figure 2 illustrates the entity-relationship diagram for RentSyne, showing the entities, their attributes, and relationships between them. The ERD helps in understanding the data model and how different entities interact with each other.

7.3. Timeline (Gantt Chart)

The base timeline for the development of RentSyne is as follows,

Task	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9	Week 10	Week 11	Week 12
Requirements & UI Mockup	✓		✓					
App structure and navigation design		✓	✓					
Basic UI and firebase/ backend integration			✓	✓				
Admin panel design with role-based access control			✓	✓	✓			
Fetch data realtime from db						✓		
UI Polish & Documentation						✓	✓	
Final Testing & Deployment								✓

Table 1: Development Timeline of RentSyne

The timeline is divided into 12 weeks, with specific tasks allocated to each period which describes an approximate timeline for the whole development process.

7.4. UI/UX Design

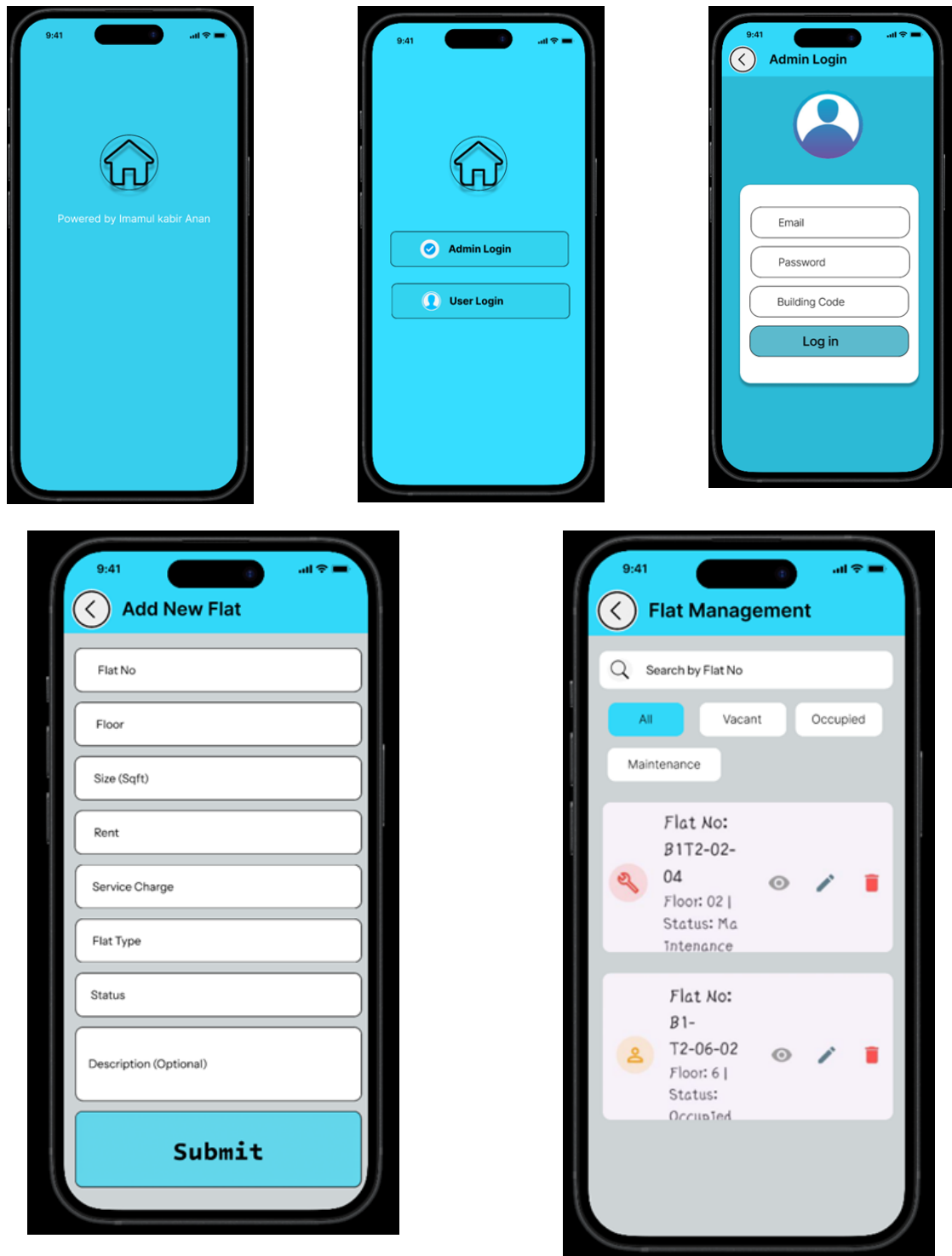


Figure 3: UI Mockups

7.5. App screens

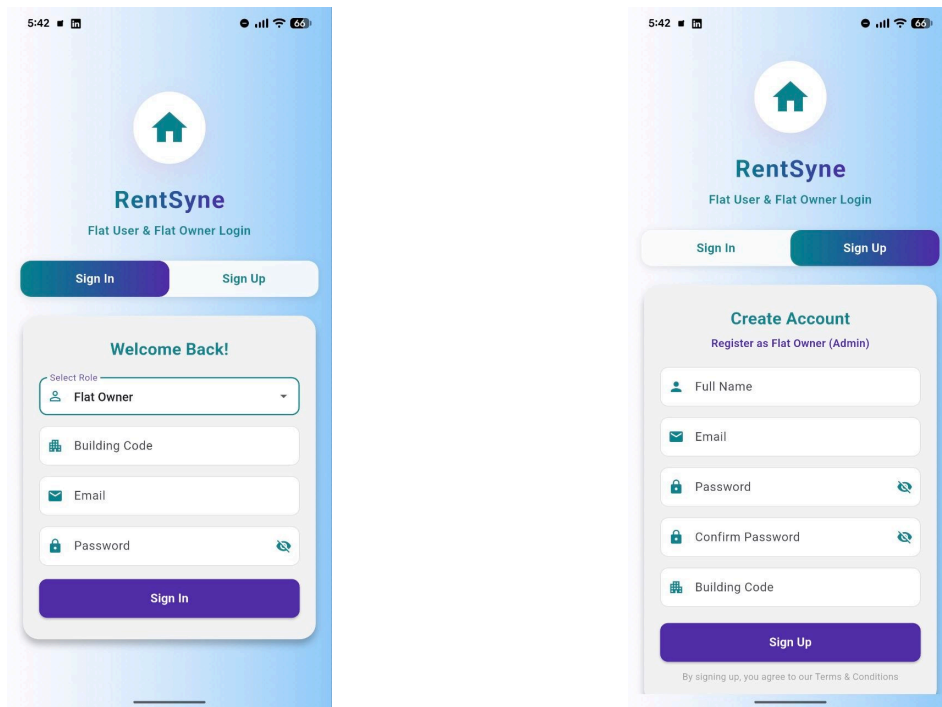


Figure 4: Sign in and Sign up screen

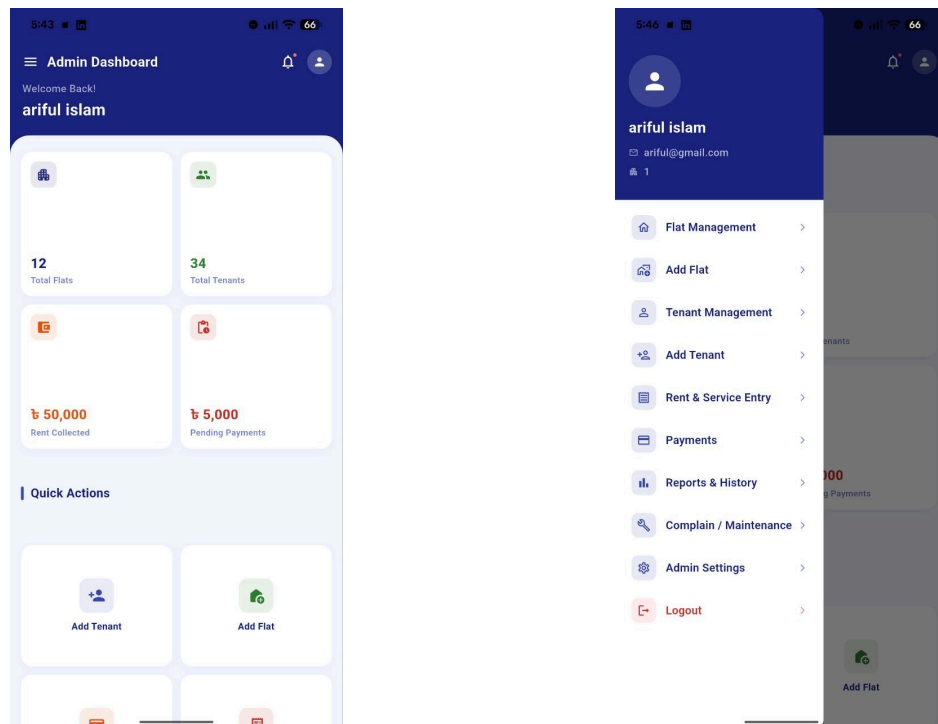


Figure 5: Admin dashboard screen

The figure shows two mobile app screens. The left screen, titled 'Add New Flat', has a dark blue header with a back arrow and a plus icon. Below the header, it says 'Fill in the flat details below'. The form is divided into sections: 'Basic Information' with fields for Flat No, Floor, and Size (sqft); 'Financial Details' with fields for Rent and Service Charge; 'Classification' with dropdowns for Type (set to 'Btg') and Status (set to 'Vacant'); and 'Additional Notes' with a text area for Description (Optional). The right screen, titled 'Add New Tenant', also has a dark blue header with a back arrow and a plus icon, with the instruction 'Fill in all required fields'. It features a 'Personal Information' section with fields for Full Name, Phone Number, Email (Optional), NID Number, Present Address, and Permanent Address, plus a checkbox for 'Same as Present Address' and two 'Emergency Conta...' fields. Below this is a 'Flat & Rent Details' section with fields for Flat No, Rent Per Month, and Service Charge.

Figure 6: Add new flat and add new tenant screen

The figure shows two mobile app screens. The left screen, titled 'Payment Accounts', has a dark blue header with a back arrow and a plus icon, with the instruction 'Manage your payment methods'. It displays three status indicators: '0 Total', '0 Active', and '0 Inactive'. Below these is a yellow warning banner that says 'No payment account added. Please add it'. A button labeled 'Add Payment A/C' with a right arrow is present, with the text 'bKash, Nagad, Rocket, Bank & more' below it. The right screen shows the 'Add Payment A/C' modal form. It has a dark blue header with a close icon. The form includes a dropdown for 'Account Type' (set to 'bKash'), fields for 'Account Holder Name *' and 'Mobile Number *', a checkbox for 'Extra Notes (Optional)', and a 'Status' dropdown (set to 'Active'). At the bottom are 'Cancel' and 'Save Account' buttons.

Figure 7: Payment account and add new payment account screen

8. Limitations

1. Single Building Support Only

The current version supports only one apartment/building per admin. Multi-building or property group management is not yet included.

2. **Limited Language Support**

Currently available in one language (e.g., English or Bangla). Multi-language functionality is not implemented.

3. **Basic Notification System**

Push notifications are implemented, but not fully customizable by admin (e.g., scheduling or template editing).

4. **User Role is Fixed**

A tenant cannot be assigned multiple roles or change roles dynamically without admin intervention.

9. **Future Plans**

1. **Multi-Building Management**

Extend the system to support multiple buildings/properties under a single admin account, allowing property managers to oversee several apartment complexes from one dashboard.

2. **Multi-Language Support**

Implement internationalization (i18n) to support multiple languages including English, Bangla, and other regional languages for broader accessibility.

3. **Advanced Notification System**

Enhance push notifications with customizable templates, scheduled reminders, and admin-configurable notification rules for rent due dates, service charges, and announcements.

4. **Expense Tracking & Reports**

Add features for tracking building maintenance expenses, generating monthly/yearly financial reports, and expense categorization for better financial management.

5. **Tenant Portal Enhancement**

Implement tenant complaint/request system, maintenance tracking, document upload (lease agreements), and community bulletin board.

6. **Analytics Dashboard**

Provide visual analytics for rent collection rates, payment trends, occupancy statistics, and financial forecasts for data-driven decisions.

7. **Web Admin Panel**

Develop a comprehensive web-based admin panel for easier management on desktop/laptop alongside the mobile app.

10. **Result**

Despite the limitations, the system achieves significant success in solving real-life problems of rent and service charge management:

1. **Successful Rent and Service Charge Tracking**

Admins can now assign, track, and manage rent and utility bills in a structured digital format.

2. **Transparent Payment History**

Tenants can view their past payments, outstanding dues, and service breakdowns — improving clarity and reducing conflicts.

3. Role-based Access Control Works Efficiently

Admins and tenants access only the features relevant to them, ensuring usability and security.

4. Digital Receipt Generation Saves Time

Automatic receipt generation after payments saves time and provides proof of transaction.

5. Complaint Management is Centralized

Tenants can easily submit complaints, and admins can update and track complaint resolution status.

6. Admin Dashboard Summarizes Monthly Overview

Collection totals, due reports, and payment stats are summarized for easy monthly review.

11. References

- [1] "Rentredi - USA Version." Accessed: Jul. 18, 2025. [Online]. Available: <https://rentredi.com/>
- [2] "Baribodol ." Accessed: Jul. 18, 2025. [Online]. Available: <https://baribodol.com/>
- [3] "Flutter." [Online]. Available: <https://flutter.dev/>

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